Stochastic Dynamics of Polymarket Contract Probabilities

Evidence for Duration-Dependent Marked Jumps

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Abstract

This paper studies how recorded probabilities evolve in binary Polymarket contracts. The main sample contains 803,862 hourly observations from 562 resolved markets associated with 157 real-world events. The data differ sharply from the increments implied by a continuous Gaussian diffusion. Among regular interior observations, 62.40% of one-hour changes are exactly zero; nonzero changes are heavy-tailed; adjacent nonzero changes reverse sign 62.21% of the time; and the probability of another change falls from 68.24% immediately after a change to 1.72% after at least 24 quiet hours. These facts motivate a hurdle model that separates the arrival of a price change from its direction and magnitude. A state-, duration-, and path-conditioned marked jump model reduces event-weighted negative log likelihood by 7.74% relative to a state-only jump model in development data. Applied without refitting to a later cohort, it improves the same score by 7.88% and wins all 27 eligible event-level comparisons. The arrival component also improves event-weighted Brier score from 0.261 to 0.174 in the later cohort. However, a separately frozen protected implementation is rejected because it produces legal forecasts for only 65 of 100 scheduled decisions. The evidence therefore supports duration-dependent marked jumps as a useful model of recorded prediction-market prices, but it does not establish an executable trading strategy.

Keywords: prediction markets; Polymarket; jump processes; duration models; probability forecasting; temporal validation.

1 Introduction

A binary prediction-market contract pays one dollar if a specified event occurs and zero otherwise. A YES contract trading at \$0.65 is therefore commonly described as assigning a 65% probability to the event. That shorthand is useful, but the observed price is not the same object as the true event probability. It is produced by orders, trades, liquidity, market rules, and a particular data feed. Interpreting it requires both probability-forecasting theory and market microstructure [7, 6, 8].

This distinction matters when choosing a time-series model. An ordinary financial asset can continue trading indefinitely and is often approximated locally by a continuous return process. A binary contract is bounded between zero and one and terminates at one of those values. New information may arrive irregularly, trading can be inactive, and the recorded series can remain unchanged for long periods. A model that treats every hour as a small draw from a smooth distribution may consequently mistake the mechanics of information arrival for low volatility.

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The paper asks two questions. First, which elements of standard financial time-series analysis remain useful for prediction-market contracts once boundedness, terminal resolution, and irregular updating are respected? Second, which stochastic-process family best describes and predicts the recorded probability series? The analysis is deliberately narrower than a claim of market inefficiency. Predicting whether a price will update is not equivalent to predicting its direction, and neither result is automatically tradable after spreads, fees, depth, and latency.

Four findings emerge. First, recorded probabilities are dominated by exact zeros and occasional large changes. Second, update timing has strong duration dependence: long inactivity predicts continued inactivity, while recent movement predicts further movement. Third, a zero-inflated marked-jump model that conditions on state, quiet duration, and recent path history predicts a later event cohort better than both a bounded Gaussian comparator and simpler jump models. Fourth, the strongest formal protected system fails its predeclared coverage requirement even though its completed forecasts are reasonably accurate. The last result is important because it separates a valid empirical regularity from a deployable forecasting system.

The contribution is methodological and descriptive. It supplies a reproducible empirical characterization of Polymarket probability paths, tests a model family on genuinely later events, and records the failure of a protected implementation rather than discarding it. The complete research state, including corpus identities, feature accesses, failures, and result receipts, is retained by the Chronos research harness.

2 Institutional and Statistical Setting

2.1 What the recorded probability represents

Let $P_t \in [0, 1]$ denote the registered YES probability at observation time t. Three quantities should not be conflated:

1. the objective probability that the event will occur;
2. the price at which a trader can execute a new order; and
3. the probability recorded by the historical data source.

This paper observes the third quantity. Depending on the venue and endpoint, a historical sample may reflect recent trades, a quote-derived value, rounding, or a stale observation. We do not assume that P_t is an unbiased estimate of the objective probability or an executable midpoint. The paper instead asks which law describes changes in the recorded series and whether that law transfers to later contracts.

2.2 Why a smooth diffusion is an incomplete default

For a continuous diffusion, an increment over a short interval is continuously distributed and therefore equals zero with probability zero. Its variance typically grows approximately in proportion to elapsed clock time. A binary prediction-market series need not satisfy either property. It can remain exactly unchanged until a trade, quote update, or information event moves the recorded probability. Its local behavior can also depend strongly on its position near zero or one and on time remaining until resolution.

The appropriate comparison is between different observation laws, not between efficient and inefficient markets. A martingale restriction can remain a useful benchmark for conditional direction

while the arrival of observations follows a duration process and nonzero increments follow a heavy-tailed marked distribution. Duration models such as the autoregressive conditional duration framework [3] and self-exciting point processes [4, 1] provide more natural precedents for clustered activity than a constant-rate diffusion.

3 Data and Research Design

3.1 Samples

The main development sample contains 562 resolved markets belonging to 157 event groups and 803,862 hourly probability observations. Markets are grouped by the real-world event they share; for example, several related candidate or outcome contracts may belong to one election event. A later cohort contains 147 markets from 29 wholly new event groups. A separate official-history panel contains five-minute observations for ten markets selected before that check was run. Table 1 summarizes the evidence sets.

Table 1: Evidence sets and their roles

Sample	Events	Markets	Observations	Role in the analysis
Main development cohort	157	562	803,862	Description and chronological model development
Later event cohort	29	147	42,549[a]	Evaluation of the frozen process law without refitting
Official five-minute panel	10	10	19,258	Independent check on sampling and source mechanics
Formal arrival schedule	111	–	231 decisions	Governed evaluation of the arrival model

Notes: [a]Observations used in the later process-signature replay. The formal arrival schedule is decision-based and is not an additional independent price corpus. Exact corpus identities and partitions are reported in Appendix B.

3.2 Temporal and event-level controls

The design enforces four rules. First, every feature must be available at the forecast timestamp; future observations cannot enter labels, transformations, or fitted parameters. Second, whole events remain together when data are partitioned. This prevents closely related contracts from the same event appearing on both sides of an evaluation boundary. Third, aggregate scores give each event equal total weight, so a long or densely sampled event does not dominate the result. Fourth, a frozen system is scored on all declared decisions, including decisions for which it cannot legally produce a forecast.

Three notions of time are kept separate: row order, elapsed UTC time, and time remaining until market close. Headline one-hour increments require an elapsed interval between 45 and 75 minutes. Twenty-four-hour variance calculations use observations whose actual gap is close to 24 hours. This prevents irregularly spaced samples from being treated as equally spaced returns.

The process law is developed on the earlier events, frozen, and then replayed unchanged on the later cohort. The separately protected arrival system has an additional fixed schedule, training

window, feature set, scoring rule, and failure policy. Once exposed, that protected schedule is not reused as a holdout.

4 Empirical Properties of the Probability Paths

4.1 Zeros, tails, reversals, and boundaries

Table 2 reports five features of the hourly series. The most immediate fact is the frequency of exact zeros. Among regular interior observations, 62.40% of one-hour increments are unchanged. A continuous Gaussian law cannot assign positive probability to that outcome. Conditional movement is also far from Gaussian: the raw kurtosis of the increment distribution is 157.10, compared with 3 under a Gaussian distribution.

The 24-hour variance ratio compares the variance of a 24-hour change with 24 times the variance of a one-hour change. A ratio near one is consistent with independent increments of constant local variance. The pooled estimate is 0.235, indicating strongly sublinear clock-time scaling. In addition, 62.21% of consecutive nonzero moves have opposite signs. This reversal rate may reflect quote bounce, order-flow correction, discrete repricing, or genuine short-run reversal; without executable bid and ask data it should not be interpreted as a trading return.

Table 2: Empirical signatures of recorded hourly probability changes

Property	Estimate	Interpretation
Exact zero increments	62.40%	The observation law contains a point mass at zero.
Raw increment kurtosis	157.10	Large changes occur much more often than under Gaussian scaling.
Pooled 24-hour variance ratio	0.235	Variance grows substantially less than linearly in clock time.
Reversal after a nonzero move	62.21%	Adjacent nonzero directions are not independent.
Persistence near the boundaries	98.99% low; 97.07% high	Behavior differs near zero and one.

4.2 Duration dependence

The probability of a future update depends strongly on how long the series has already remained unchanged. Figure 1 groups observations by the current quiet run. A market that has just moved changes again at the next regular hourly sample 68.24% of the time. After one to five quiet hours, the rate falls to 28.90%; after six to 23 quiet hours, it is 9.50%; after at least 24 quiet hours, it is only 1.72%.

This pattern rejects a constant-rate Poisson arrival clock. It also differs from the intuition that an event becomes overdue after a quiet spell. In these data, activity clusters: recently active markets tend to remain active, while inactive markets tend to remain inactive.

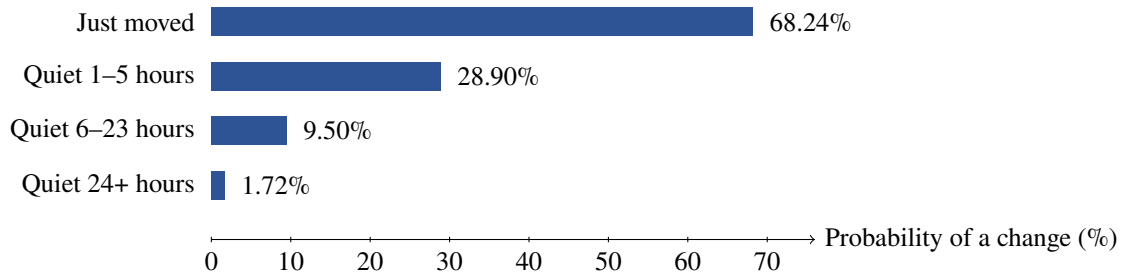


Figure 1: Observed probability of a nonzero change at the next regular hourly sample, grouped by the duration of the current zero run. Long quiet spells predict continued inactivity rather than an increasing chance of an imminent move.

4.3 Independent source and frequency check

This is a narrow validation exercise, not the main empirical sample. The main estimates above use 562 markets from 157 events, and the unchanged later-cohort evaluation uses another 147 markets from 29 events. The ten-market panel is used only to ask whether the main signatures disappear when the same contracts are observed through an independently acquired official source at a five-minute frequency.

Hourly sampling could create artificial zeros or reversals if a finer path were moving between the recorded points. The official five-minute panel therefore provides a targeted check. On that panel, 78.00% of changes are zero, kurtosis is 203.64, the reversal rate is 67.87%, and the 24-hour variance ratio is 0.067. Of 1,580 hourly observations aligned to an official-history sample within ten minutes, 99.49% match exactly.

These results show that coarse hourly sampling alone does not generate the reported signatures on the selected panel. They do not identify the economic source of the zeros. Inactivity, stale quotes, price rounding, bid-ask mechanics, and publication rules can all contribute to the recorded process.

5 Models and Evaluation

5.1 Comparator and marked-jump specification

Let $\Delta P_{t+1} = P_{t+1} - P_t$ and let $\mathcal{F}_t$ denote information legally available at time t. The continuous comparator is a zero-mean Gaussian increment with variance that shrinks near the boundaries:

$$\Delta P_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}\Big(0, \sigma^2(c_t) P_t (1 - P_t)\Big), \tag{1}$$

where c_t is a fixed state and time-to-close cell. This is a deliberately simple bounded diffusion benchmark, not a claim that all continuous-time latent-price models reduce to Equation (1).

The alternative is a hurdle or zero-inflated marked-jump model. It decomposes the next increment into three questions: whether the recorded price changes, the sign of a nonzero change, and the magnitude of that change:

$$\Pr(\Delta P_{t+1} = 0 \mid \mathcal{F}_t) = 1 - q_t, \tag{2}$$

$$\Pr(\Delta P_{t+1} > 0 \mid \Delta P_{t+1} \neq 0, \mathcal{F}_t) = s_t, \tag{3}$$

$$\log |\Delta P_{t+1}| \mid \Delta P_{t+1} \neq 0, \mathcal{F}_t \sim \mathcal{N}(\mu_t, \tau_t^2). \tag{4}$$

Four nested specifications are compared. The memoryless model uses global arrival, sign, and size laws. The state-conditioned model uses the current probability and time to close. The full model adds the current quiet duration, recent inactivity, the previous nonzero sign, and the previous nonzero magnitude. Cells shrink toward an event-weighted global law to reduce instability in sparse regions. Appendix A gives the exact interval-probability score.

5.2 Evaluation metrics

The full observation law is evaluated by negative log likelihood (NLL). For each observed increment, the model is penalized by the negative logarithm of the probability it assigned to the observed zero or to a common-width interval around the observed nonzero change. Lower NLL is better. Observation-weighted NLL answers how well the model fits a randomly selected row; event-weighted NLL first averages within events and then weights each event equally.

The arrival component predicts $Y_{t+1} = \mathbb{1}[\Delta P_{t+1} \neq 0]$ and is evaluated by the Brier score,

$$\mathrm{BS} = \frac{1}{n} \sum_{i=1}^{n} (\widehat{q}_i - Y_i)^2, \tag{5}$$

where zero is perfect and smaller values indicate better probability forecasts [2]. Calibration, mean absolute error, coverage, and failure concentration are also reported where appropriate.

6 Results

6.1 Predictive density of the full process

Table 3 compares the process models. In the development cohort, the full marked-jump model obtains event-weighted NLL of 1.941, compared with 2.104 for the state-only jump model and 3.578 for bounded Gaussian diffusion. The improvement over the state-only jump law is 7.74%, and the full model wins 112 of 113 eligible event comparisons.

The model is then frozen and applied without refitting to the later cohort. Its event-weighted NLL is 2.870, compared with 3.116 for the state-only jump model, an improvement of 7.88%. It wins all 27 eligible later-event comparisons against both the state-only model and the diffusion comparator. The absolute NLL is worse in the later cohort than in development. Thus the structural form transfers, while the parameter levels do not remain perfectly calibrated.

Table 3: Negative log likelihood by process specification

Model	Development observation NLL	Development event NLL	Later cohort event NLL
Bounded state-dependent Gaussian diffusion	3.511	3.578	4.582
Memoryless zero-inflated marked jump	1.865	2.214	3.240
State-conditioned zero-inflated marked jump	1.782	2.104	3.116
State-, duration-, and path-conditioned marked jump	**1.585**	**1.941**	**2.870**

Notes: Lower values are better. Event NLL gives every real-world event equal weight. The later-cohort values are produced by applying the frozen development law without refitting.

6.2 Forecasting whether the price will change

The clearest predictive component is the arrival model. A regularized logistic regression uses the current probability, time to close, the logarithm of the current quiet run, and the fraction of zero increments in the preceding 24 hours. On the later cohort, adding the inactivity variables reduces event-weighted Brier score from 0.261 for a state-only model to 0.174, a 33.38% improvement. The inactivity model wins all 27 eligible event-level comparisons. Its average forecast probability is 41.29%, while the observed update rate is 45.34%, for an absolute calibration gap of 4.05 percentage points.

This is a forecast of whether the *recorded series changes*, not a forecast of whether the contract is underpriced or overpriced. A movement forecast does not specify direction, expected profit, or the price at which a position can be entered.

6.3 Protected evaluation and rejection

After development, the continuous arrival model was frozen together with its source code, features, training requirements, decision schedule, metrics, and failure rules. It was evaluated once on 100 protected decisions. The system produced 65 forecasts and failed to produce 35. On the completed forecasts, Brier score was 0.189 and mean absolute error was 0.394, both within the predeclared limits. Coverage and failure rate were not.

Table 4: Protected evaluation of the frozen arrival system

Criterion	Result	Required	Decision
Brier score on completed forecasts	0.189254	≤ 0.200000	Pass
Mean absolute error on completed forecasts	0.394489	≤ 0.400000	Pass
Coverage	0.650000	≥ 0.950000	Fail
Failure rate	0.350000	≤ 0.050000	Fail
Maximum error concentration	0.054852	≤ 0.050000	Fail

Notes: Thirty-four decisions lacked the fixed legal training history and one lacked a legally available outcome. The system is rejected under the predeclared joint criteria.

The protected result cannot be rescued by reporting only the completed rows. Missing forecasts may be systematically more difficult than completed forecasts, so conditional accuracy does not establish system-level validity. The rejection does not undo the earlier observation that quiet duration predicts update timing. It prevents the exact frozen implementation from being promoted as a validated forecasting system.

7 Implications for Financial Modelling and Trading

7.1 Which financial ideas transfer

Several established ideas remain useful. Chronological validation, state-dependent volatility, heavy-tailed conditional returns, duration models, and point-process methods all transfer, but they require prediction-market-specific modifications. The natural unit of independence is the real-world event rather than the individual price row. The state space is bounded, resolution is terminal, and the no-change probability must be modelled separately from conditional magnitude.

Other defaults transfer poorly. A constant Brownian diffusion cannot represent an empirical point mass at zero or the observed duration dependence. Standard Black–Scholes replication has no direct analogue when there is no continuously traded underlying asset that replicates the event payoff. Martingale reasoning remains a useful directional benchmark, but it must distinguish the recorded historical probability from an executable price, a latent belief, and the physical event probability.

7.2 Why the result is not yet a trading strategy

The process evidence is strongest for timing and weakest for executable direction. Separate internal-data and external-data studies have not yet cleared the required controls. A raw 15-minute fill-flow association of approximately -0.46 fell to about $+0.04$ to $+0.05$ after maker and taker legs were converted to a consistent transaction direction. This correction was necessary because the retained fill table is a per-party ledger rather than a one-row-per-trade record.

A separate study reconstructed 31 publication-time-legal Brazilian presidential runoff polls. These releases yielded 25 usable decisions and 17,570 canonical Polymarket fills. Follow and fade rules were evaluated at several thresholds and horizons. Every rule with at least five executable pairs was negative after conservative fees. The broad 72-hour fade produced \$10.69 before costs and $-$\$18.97 after costs over seven trades. These results are too small and too sparse to support a general trading conclusion, but they illustrate why predictive association and net trading profit are different standards.

No Sharpe ratio or return claim is reported for the main process model because the required execution evidence is absent. A credible trading test would need side-specific executable quotes, spread, fees, depth, latency, order-filling assumptions, and an abstention rule fixed before the protected evaluation.

8 Limitations and Research Agenda

8.1 Scope of the evidence

The primary evidence is broad. The development sample contains 562 markets associated with 157 events, and the later evaluation adds 147 markets associated with 29 new events. The ten-market five-minute panel serves a different purpose. It is a targeted robustness check using an independently acquired source: it asks whether the principal hourly signatures survive at a higher sampling frequency. The paper should therefore be read as a broad study of registered hourly probability paths, supported by a narrower high-frequency check. It is not a venue-wide study of five-minute order books.

8.2 What remains unidentified

The main data source records a probability at each timestamp, but it does not establish whether that value is a last trade, a midpoint, another quote-derived measure, or a stale published value. Consequently, an unchanged observation may represent genuine inactivity, an unchanged quote, or the behavior of the historical feed. Separate studies in Chronos use canonical transaction records, but those fills cannot retrospectively identify every observation in the main corpus. Distinguishing economic inactivity from publication mechanics requires historical trades and order books observed together.

Two further qualifications concern statistical scope. Keeping related contracts within the same event prevents the most direct form of leakage, but separate events may still respond to common political or economic information. In addition, the Gaussian comparator tests a simple bounded diffusion; rejecting it does not exclude every continuous latent-price model with endogenous sampling or market-microstructure noise.

The validation evidence must also be interpreted precisely. The later cohort tests whether a frozen model family transfers to new events, but it is not a second protected experiment. The one protected implementation was rejected because it failed to produce enough legal forecasts. That schedule is now exposed and cannot be reused as an untouched holdout.

8.3 Research agenda

The next experiment should join transactions, executable quotes, and order-book depth in a common event-time panel. Such data would make it possible to separate trade arrival from quote updates, measure bid-ask bounce and price impact directly, and determine whether apparent reversals remain after realistic execution costs.

The modelling framework can retain the useful decomposition developed here. A first component estimates whether an economically meaningful update is likely to occur; a second predicts its direction; and a third predicts its magnitude. Directional inputs may include time-legal polling releases, macroeconomic announcements, and prices from related prediction venues. These variables are candidate predictors, not assumed arbitrage relationships. Any specification that passes chronological development must be frozen and evaluated on a new event cohort using executable bid and ask prices, fees, depth, latency, and a predeclared abstention rule.

9 Conclusion

Recorded Polymarket probabilities do not behave like smooth Gaussian diffusions sampled once per hour. They exhibit a large point mass at zero, extreme conditional changes, reversal in adjacent nonzero moves, sublinear variance scaling, and strong dependence on the duration of inactivity. A marked-jump model that separates arrival, direction, and magnitude fits these features directly. Its advantage over state-only and Gaussian alternatives transfers to a later cohort without refitting, and its arrival component materially improves probability forecasts of whether the next recorded sample will change.

The evidence supports a stochastic description, not a profitable strategy. The exact protected arrival system failed its coverage requirement, and directional studies have not survived execution and temporal controls. The defensible conclusion is therefore specific: duration- dependent marked jumps provide a better working model for recorded Polymarket probability paths, while tradable directional prediction remains an open empirical problem.

A Exact Scoring Definitions

For a nonzero observed increment x, every model assigns probability to the same interval of width 0.001. Under the marked-jump model this probability is

$$q(\mathcal{F}_t)\Pr(\operatorname{sign} x \mid \mathcal{F}_t)\left[\Phi\left(\frac{\log(|x|+\epsilon)-\mu}{\tau}\right)-\Phi\left(\frac{\log(\max\{10^{-6},|x|-\epsilon\})-\mu}{\tau}\right)\right], \tag{6}$$

where $\epsilon = 0.0005$. A zero increment receives point mass $1 - q(\mathcal{F}_t)$. The reported NLL is the mean negative logarithm of this probability, calculated both by observation and after averaging equally across events.

The continuous arrival model is

$$\operatorname{logit}\Pr(Y_{t+1}=1 \mid \mathcal{F}_t) = \beta_0 + \beta_p f(P_t) + \beta_T f(T_t) + \beta_q \log(1+Q_t) + \beta_z Z_{t,24}, \tag{7}$$

where T_t is time to close, Q_t is the current quiet run, and $Z_{t,24}$ is the fraction of zero increments during the preceding 24 hours. Transformations and coefficients are fitted only on legally prior data.

B Data Authority and Corpus Identity

The opening 1,000-contract corpus contains 235 events and 1,180,437 observations. The successor acquisition contains 1,509 contracts, 321 events, and 1,585,740 observations. Its content identity is:

```
polymarket-corpus:
184b83f7f8012da4ca9116c8b6bad2076570ad5f0bada95b22615ee2f5cceae8
```

The successor partition contains 29 wholly fresh development events comprising 147 markets, 227 embargoed ancestor events comprising 1,005 markets, and 65 protected events comprising 357 markets. Data were acquired through the read-only `human_ro` database role. Chronos retains source digests, selections, temporal semantics, partitions, actual field accesses, worker outcomes, and result receipts.

C Robustness Across Representations and Event Strata

Table 5 tests whether the main signatures are confined to boundary values, a particular coordinate system, or a maturity band. Exact zero mass and high kurtosis remain in the strict interior and after transforming probabilities to log odds. The event-equal 24-hour variance ratio remains below one in every fixed view.

Table 5: Representation and maturity robustness

Fixed view	Increments	Zero rate	Kurtosis	Event-equal 24h ratio
All probability rows	781,388	69.73%	157.10	0.358
Strict interior probability	519,495	62.76%	103.19	0.376
Strict interior log odds	519,495	62.76%	104.99	0.403
More than 168 hours to close	701,428	69.40%	151.06	0.338
Zero to 168 hours to close	79,960	72.54%	138.44	0.512

Table 6 shows event-equal estimates across contract density, lifetime, and history length. Parameter magnitudes vary, but high zero mass, sublinear variance scaling, excess reversal, and lower movement rates following quiet runs persist across the reported strata.

Table 6: Structural estimates across whole-event strata

Whole-event stratum	Zero mass	Variance ratio	Reversal	Quiet ratio
One-market events	0.577	0.403	0.588	0.084
At least five markets	0.527	0.267	0.598	0.032
Lifetime at most 30 days	0.477	0.293	0.581	0.038
Lifetime 31–180 days	0.605	0.425	0.594	0.078
Lifetime more than 180 days	0.623	0.228	0.643	0.035
Fewer than 500 points	0.458	0.315	0.556	0.014
500–1,999 points	0.586	0.394	0.601	0.070
At least 2,000 points	0.631	0.248	0.635	0.053

D Claim Boundaries

Table 7: Status and boundary of the main empirical claims

Claim	Status	Boundary
Smooth constant-intensity diffusion describes registered samples	Rejected	Zero mass, tails, scaling, and duration dependence contradict the comparator.
State-, duration-, and path-conditioned jump law improves predictive density	Supported in development and later replay	The result targets recorded samples and is not a protected trading finding.
Inactivity improves update prediction	Supported in development and later replay	The exact protected implementation was rejected.
Boundaries alone explain stickiness	Rejected as a sole explanation	Interior-probability and log-odds checks retain the signatures.
Hourly sampling alone creates the signatures	Rejected on the selected panel	The independent check contains only ten markets.
Sign reversal is executable mean reversion	Not established	Executable quotes, spread, fees, depth, and latency are absent.
Chronos has identified tradable alpha	No	Cost-aware directional ideas were rejected or remain unvalidated.

Data and Code Availability

Numerical claims map to content-addressed receipts in `paper/evidence-index.md`. Replay commands and the non-rerunnable protected boundary are documented in `paper/reproducibility.md`. The manuscript is a presentation layer and does not alter the epistemic status of any underlying result.

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